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# Assessment of drought trends and variabilities over the agriculture-dominated Marathwada Region, India

**Sabyasachi Swain · Surendra Kumar Mishra · Ashish Pandey · Deen Dayal**

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**Abstract** Drought is considered among the most perilous events with catastrophic consequences, particularly from the agro-economic point of view. These consequences are expected to exacerbate under the increasing meteorological aberrations due to changing climate, which necessitates investigating drought variabilities. This study presents a thorough spatiotemporal assessment of drought trends and variabilities over the agriculture-dominated Marathwada Region, Maharashtra, India. The precipitation data is extracted from the India Meteorological Department (IMD) gridded product, whereas actual evapotranspiration (ET) and Evaporative Stress Index (ESI) are obtained from Global Land Evaporation Amsterdam Model (GLEAM) datasets. Standardized Precipitation Index (SPI) is used to characterize drought occurrences at multiple time frames, whereas non-parametric tests, i.e., modified Mann–Kendall (MMK) and Sen's slope (SS) tests, are employed to detect trends. The results reveal the region to be prone to droughts, and SPI at a longer time frame (i.e., 12-monthly moving frame) can capture drought occurrences better than the shorter time frames, which can be attributed to the lesser randomness in the time series in the longer frame. A mix of positive/negative trends of SPI series are found for the monsoonal months; however, they are relatively more concentrated towards negative $Z_{MMK}$. Hence, the Marathwada Region can be inferred to have exhibited a relatively increased tendency towards drought occurrences. The seasonal differences in mean values and trends of rainfall, ET, and ESI are discussed in detail. Since the Marathwada Region has a monsoon-dominated climate with high agricultural importance, the information reported in this study will help in devising water management strategies to minimize the repercussions of droughts.

**Keywords** Meteorological drought · Marathwada · Drought trend · SPI · GLEAM · ESI

## Introduction

Climate change has far-reaching impacts on agriculture. This is due to the agricultural system's reliance on the climate and the intricate roles that agriculture plays in socio-economic systems at regional, national, and global levels. Climate change can have both positive and negative effects on the productivity, location, and timing of the plant/crop systems at all these levels (Behera & Kushwaha, 2012; Das & Behera, 2019; Swain et al., 2021a). In addition, as the world strives to feed over a population of 8.5 billion

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S. Swain (✉) · S. K. Mishra · A. Pandey · D. Dayal
Department of Water Resources Development and Management, Indian Institute of Technology Roorkee, Roorkee 247667, India
e-mail: sswain@wr.iitr.ac.in

by 2030, climate change will pose new food security challenges and hamper the stability of food supplies (Zhang, 2008). Furthermore, the agricultural sector contributes to the global economy and trade, specifically in terms of import and export of food items and other agricultural products. The amount of produce available for import and export as well as its pricing will be impacted by climate change.

The rainfall time series naturally fluctuates from inter-annual or intra-annual variations. Drought is also a natural part of the hydroclimatic cycle. Drought is a climatic extreme, specifically regarded as a persistent deficit of water, which leads to detrimental impacts on agriculture and the people who directly depend on it (Afzal & Ragab, 2020; Lotfirad et al., 2021). Among all the meteorological crises, droughts have the highest deleterious impacts on the people and the economy (Amrit et al., 2018a, b; Swain et al., 2021a). Droughts are the most severe natural hazards on the planet, occurring in every region and affecting a larger area than any other hazard (Swain et al., 2021a; Wilhite et al., 2007). Recent research on hydrologic extremes shows that, despite technological advancements, every part of the world is vulnerable to the consequences of droughts (Althoff et al., 2021; Swain et al., 2020a, b, 2022a, b, c). Prediction of droughts is highly challenging because of the difficulty in estimating their beginning and end (Rossi et al., 1992; Tigkas et al., 2012; Tsakiris et al., 2013). Droughts are broadly divided into four forms, i.e., meteorological, agricultural, hydrological, and socioeconomic (Mishra & Singh, 2010; Mishra et al., 2009; Swain et al., 2022b). However, drought is mostly instigated due to persistent rainfall deficit (commonly regarded as meteorological drought), which propagates to other forms. A meteorological drought can be characterized as a lack of rainfall during a season or year at a certain location when compared to the usual seasonal or annual rainfall of that location (Tsakiris & Vangelis, 2004; Wilhite & Glantz, 1985). Drought has several pernicious impacts; i.e., it depletes the water table, reduces reservoir storage, and causes crop failure. The impacts are compounded due to the dramatic rise in water consumption in recent decades as a result of fast population growth, irrigation, urbanization, and industrialization (Wilhite et al., 2014). Moreover, water depletion and contamination are caused by the overuse of water resources and rampant anthropogenic activities across different parts of the globe (Gosain et al., 2006, 2011; Swain et al., 2021a). Under such circumstances, climatic changes and variabilities leading to flickering rainfall patterns and droughts further exacerbate the predicaments (Guptha et al., 2022; Patel et al., 2022; Singh et al., 2021; Swain et al., 2021a, b, 2022d).

India is a developing country with a huge population, the majority of which is directly dependent on agriculture. As agriculture in the Indian context heavily relies on the timing and quantity of monsoonal rainfall, the climatic system governs the country's social, economic, and environmental conditions. Apart from the direct impacts on agriculture, droughts under climate change may have a variety of implications on water availability, farmer's financial condition, food security, productivity of soil, biodiversity, and rural livelihoods (Mahanand et al., 2022; Panda et al., 2019; Roy et al., 2022; Shafique & Behera, 2017; Swain et al., 2021a, 2022b). However, droughts have occurred very frequently in India. They have prevailed over different parts of the country at least once in every 3 years, and their frequency and severity are gradually increasing (Amrit et al., 2019; Mishra & Singh, 2011; Swain et al., 2021b). The long-term pervasive droughts have hit successively with a higher frequency since the 1980s (Lester & Gurenko, 2003; Mishra & Singh, 2010). The most drought-affected states are Maharashtra, Karnataka, Uttar Pradesh, Gujarat, Odisha, Andhra Pradesh, Madhya Pradesh, and Rajasthan (Deshpande, 2008). To have a better assessment and predictive ability of droughts, extensive research on existing systems is required. The assessment of droughts is necessary for taking actions to improve its monitoring. It urges for rigorous analysis of drought occurrences over a particular region, including a thorough understanding of the governing climatic variables. However, the first step of drought assessment depends on correctly quantifying the drought phenomenon and its associated attributes. Despite the consequences and ubiquity of droughts, their accurate quantification has always been a perplexing geophysical endeavor (Keyantash & Dracup, 2002; Mishra & Singh, 2010, 2011).

The use of statistical methods for drought quantification and its trend detection has recently received a lot of interest. Typically, drought characterization is accomplished by using indices, which assemble the climate-related information to form a numerical value, thereby making the drought assessment

easier (Mishra & Singh, 2011; Mishra et al., 2015; Mukherjee et al., 2018). Standardized Precipitation Index (SPI) is the most popular meteorological drought index and is widely applied due to its numerous advantages over other indices. On the other hand, non-parametric methods, viz., modified Mann–Kendall (MMK) and Sen's slope (SS) tests, are widely used to detect trends in hydroclimatic variables. While the MMK test detects the nature of the trend (increasing/decreasing) along with its statistical significance, the SS test enables to estimate the magnitude of the trend in terms of relative change (RC) with respect to the long-term mean value (Guptha et al., 2021; Pandey & Khare, 2018; Sahoo et al., 2021; Swain et al., 2019, 2022a, c; Tiwari & Pandey, 2019). Several studies have been carried out to assess the trends of droughts and the associated variables, some of which are discussed below in brief. Chen et al. (2009) used SPI on a trimonthly frame to explore the historical trend and variability of meteorological droughts in Taiwan, and the findings revealed a considerable increase in the number of dry days and the frequency of droughts. Kumar et al. (2012) conducted a spatiotemporal assessment of SPI-based drought occurrences during 1951–2007 over different regions of India during the monsoon season and observed an increasing drought trend over a majority of them. Huang et al. (2014) used the MMK test to discern the drought patterns in the Wei River Basin, China, and obtained a drying trend in the eastern half of the basin from 1961 to 2005. Thomas et al. (2015) used the non-parametric tests and observed mostly decreasing trends in seasonal rainfall and the number of wet days over the drought-prone Bundelkhand Region, India. Mahajan and Dodamani (2015) assessed trends in rainfall and drought events over the Upper Krishna River Basin, which is frequently susceptible to severe droughts. The non-parametric tests identified significantly increasing drought trends over only a few stations. Achugbu and Anugwo (2016) investigated the trend of SPI-based droughts over Kano, Nigeria, and found an increased tendency towards droughts in some months. MMK test was employed by Amirataee and Montaseri (2017) to investigate spatiotemporal drought patterns in Iran during 1965–2014, which demonstrated a statistically significant increase in droughts across the northwestern parts. Using the MMK test, Ahmed et al. (2018) evaluated the meteorological drought trends in Pakistan and reported a rising drought frequency. Similarly, Qutbudin et al. (2019) reported a significant increase in frequency and severity of droughts in Afghanistan. From these studies, it can be inferred that trend analysis of droughts and the associated variables discerns useful information, which can be helpful for drought management.

Maharashtra, India's third largest state, is the worst affected by droughts among all the states (Swain et al., 2017). Marathwada is among the five divisions of Maharashtra, which is majorly covered by agricultural lands. The frequent drought occurrences over the Marathwada Region due to substantial year-to-year rainfall variation have led to repeated crop failure, resulting in a large number of farmer suicides in recent decades. However, no study has explicitly assessed the drought trends and variabilities over the region. Given all the aforementioned concerns, this study aims to assess the spatiotemporal trends and variabilities of droughts along with the governing variables over the Marathwada Region, Maharashtra, India. The materials and methods, results, and key conclusions are discussed in detail in the subsequent sections.

## Materials and methods

### Study area

Marathwada is one of the five divisions of Maharashtra State, India. It is located from 74° 40′ to 78° 15′ East longitudes and from 17° 35′ to 20° 40′ North latitudes. The Marathwada Region has eight constituent districts, viz., Aurangabad, Jalna, Parbhani, Hingoli, Nanded, Bid, Latur, and Osmanabad, which covers an area of about 64,600 km$^2$. The locations of the division along with the eight constituent districts are shown in Fig. 1. The climatic condition of the region is mostly dry except the monsoon season. The maximum daily temperature goes up to 43 °C in summer days, while the lowest daily temperature (~6 °C) is observed in winter nights. Relative humidity over the region is very low (35–40%) in non-monsoon seasons, which rises to about 85% during monsoon season. The mean

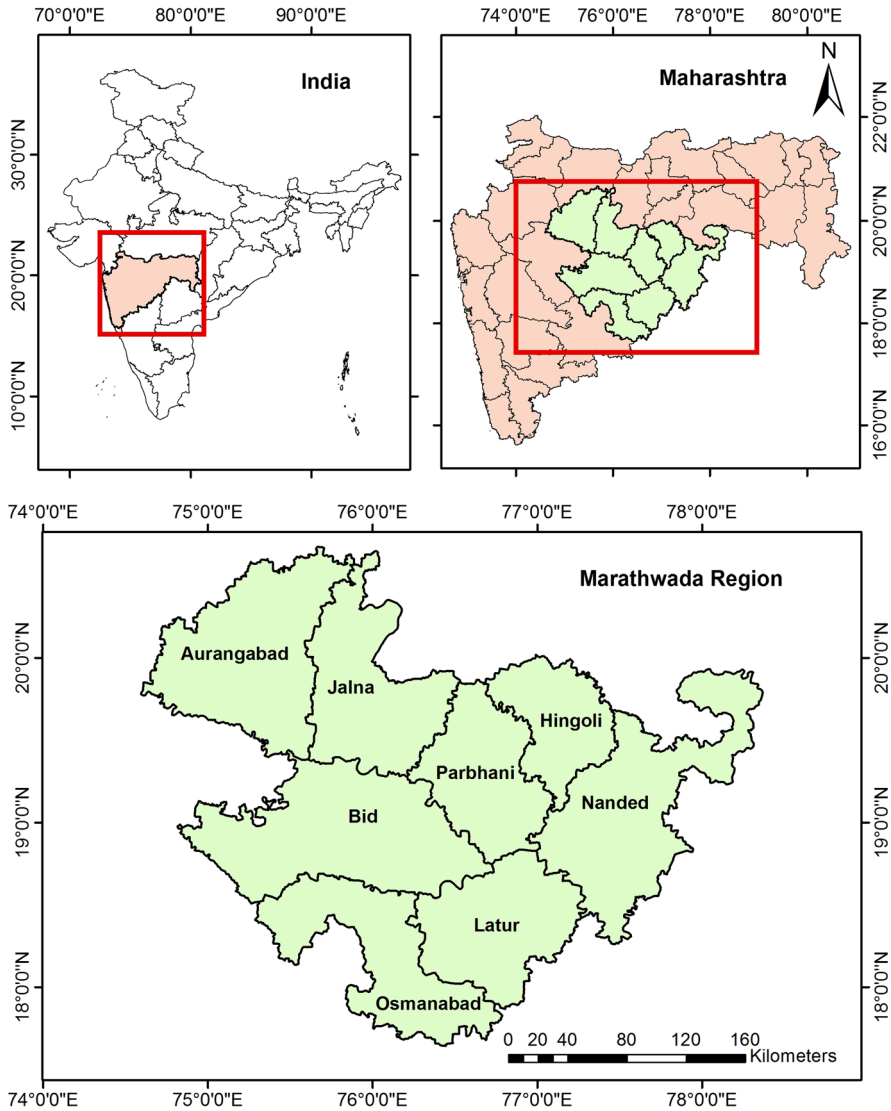



**Fig. 1** Location of Marathwada Region, Maharashtra, India, and its constituent districts

annual rainfall over the region is about 880 mm, and its 80% occurs during southwest monsoon season (June–September). The region is characterized by deep to very deep black cotton soils with good water holding capacity. Agricultural lands account for three-fourth of the total land area, which are majorly rain fed. Major crops in the region are cotton, soyabean, pigeon pea, sugarcane, wheat, bajra, and pulses. The region is prone to significant year-to-year rainfall fluctuations in monsoon season, thereby causing frequent droughts. Due to these rainfall aberrations and the consequent crop failures, Marathwada has witnessed a large number of farmer suicides in recent decades (Kulkarni et al., 2016; Swain et al., 2017).

## Datasets and methods

The data types used in this study include precipitation, actual evapotranspiration (ET), and Evaporative Stress Index (ESI). It is to mention that ESI is defined as the ratio of ET to potential evapotranspiration (PET). Precipitation is obtained from the India Meteorological Department (IMD) gridded product, which is converted into monthly scales and then arranged in multiple time frames for deriving SPI series. The

Python library 'IMDLIB' (https://doi.org/10.5281/zenodo.4405233) can be used to read and download the IMD gridded data. As ground-based observations of ET and ESI are unavailable over the study region at desirable quantity/quality, they are retrieved from the Global Land Evaporation Amsterdam Model (GLEAM), a set of algorithms that separately estimate different components of evapotranspiration. The details of the data types and their sources are presented in Table 1.

As mentioned previously, the objective of this work is to assess the spatiotemporal trends and variabilities of droughts. Therefore, the use of robust approaches for drought characterization and time series analysis are necessary. Droughts are typically characterized using indices, which assemble the climate-related information to form a numerical value, thereby making the drought assessment easier. While there are numerous indices for meteorological drought monitoring, SPI is widely accepted across the globe. Moreover, several studies on evaluation of drought indices have concluded SPI as the superior considering the overall performance attributes (Swain et al., 2021a). The World Meteorological Organization (WMO) has also recommended utilizing SPI for meteorological drought monitoring. Therefore, SPI is used in this study to characterize drought occurrences over the Marathwada Region.

SPI represents a transformed variate of the rainfall magnitude. It is computed by fitting the rainfall time series with a suitable distribution (preferably 2-parameter gamma) and then transforming the cumulative distribution function (CDF) to a reduced variate with mean of zero and standard deviation of unity. The detailed procedure to compute SPI is provided in the Supplementary Information. Besides the simple procedure, SPI possesses the advantage to be utilized in multiple time frames over diverse climatic conditions. Furthermore, classification of SPI involves uniform characterization in terms of transitioning from one severity class to another, i.e., moderate to severe or severe to extreme droughts. Typically, SPI at or below $-1$ is regarded as drought, while $-1.5 < \text{SPI} \leq -1$, $-2 < \text{SPI} \leq -1.5$, and $\text{SPI} \leq -2$ are identified as moderate, severe, and extreme droughts, respectively. The classification of SPI into different drought/wet categories is provided in Table A1 of the Supplementary Information. The probability of drought occurrence under a particular class is calculated by dividing the duration (in months) of that class by the overall duration, which can then be multiplied by 100 to obtain a percentage representation.

In this study, SPI is employed at four different moving windows, i.e., 1-, 3-, 6-, and 12-monthly time frames. The spatiotemporal variation in frequency and drought occurrence probability are analyzed. The drought trends at different time frames are also assessed for the monsoonal months, as these months cumulatively contribute a vast majority of annual rainfall. Furthermore, to develop a better understanding of meteorological drought trends and variabilities, the magnitudes and trends of meteorological variables, i.e., rainfall, ET, and ESI, are analyzed at seasonal and annual scales in detail. It is pertinent to mention that in the Indian hydroclimatic context, the water year starts from June and ends in May, which is sub-divided into four seasons, i.e., monsoon, post-monsoon, winter, and pre-monsoon seasons corresponding to June–September, October–November, December-February, and March–May, respectively.

The modified Mann–Kendall (MMK) test and Sen's slope (SS) test are employed for trend analysis of SPI time series. These two non-parametric methods are most widely used for trend analysis of hydroclimatic variables. While MMK test provides the nature of monotonicity (increasing/decreasing) of time series along with its statistical significance, SS

**Table 1** Data used in this study and details of their sources

| Data type | Details |
|---|---|
| Precipitation (1980–2020) | Source: India Meteorological Department (IMD) gridded product<br>Link: https://www.imdpune.gov.in/Clim_Pred_LRF_New/Grided_Data_Download.html<br>Resolution: 0.25° × 0.25°, daily |
| Evapotranspiration (ET), Evaporative Stress Index (ESI) (1980–2020) | Source: Global Land Evaporation Amsterdam Model (GLEAM)<br>Version: GLEAM v3.5a<br>Link: https://www.gleam.eu/<br>Resolution: 0.25° × 0.25°, daily |

test enables to estimate the magnitude of the trend in terms of relative change (RC). The detailed methodology/procedures of these two methods are provided in the Supplementary Information.

## Results and discussions

The spatial distribution of drought occurrence probability in total and in different severity categories (i.e., moderate, severe, and extreme) for four moving time frames (1-, 3-, 6-, and 12-monthly) is presented in Fig. 2. It can be observed that the probability of moderate droughts is higher than that of severe droughts, whereas the extreme droughts possess the least probability. The probability of total drought goes beyond 20% over some grids, which shows the proneness of these regions towards meteorological droughts. Interestingly, SPI at a longer time frame (i.e., 12-monthly moving frame) can capture drought occurrences better than the shorter time frames. The drought occurrence probability in case of SPI-1 is very low. This can be attributed to the randomness or fluctuations in the time series; i.e., the rainfall time series changes drastically in the 1-monthly time frame. Furthermore, the higher inter-annual variations lead to the

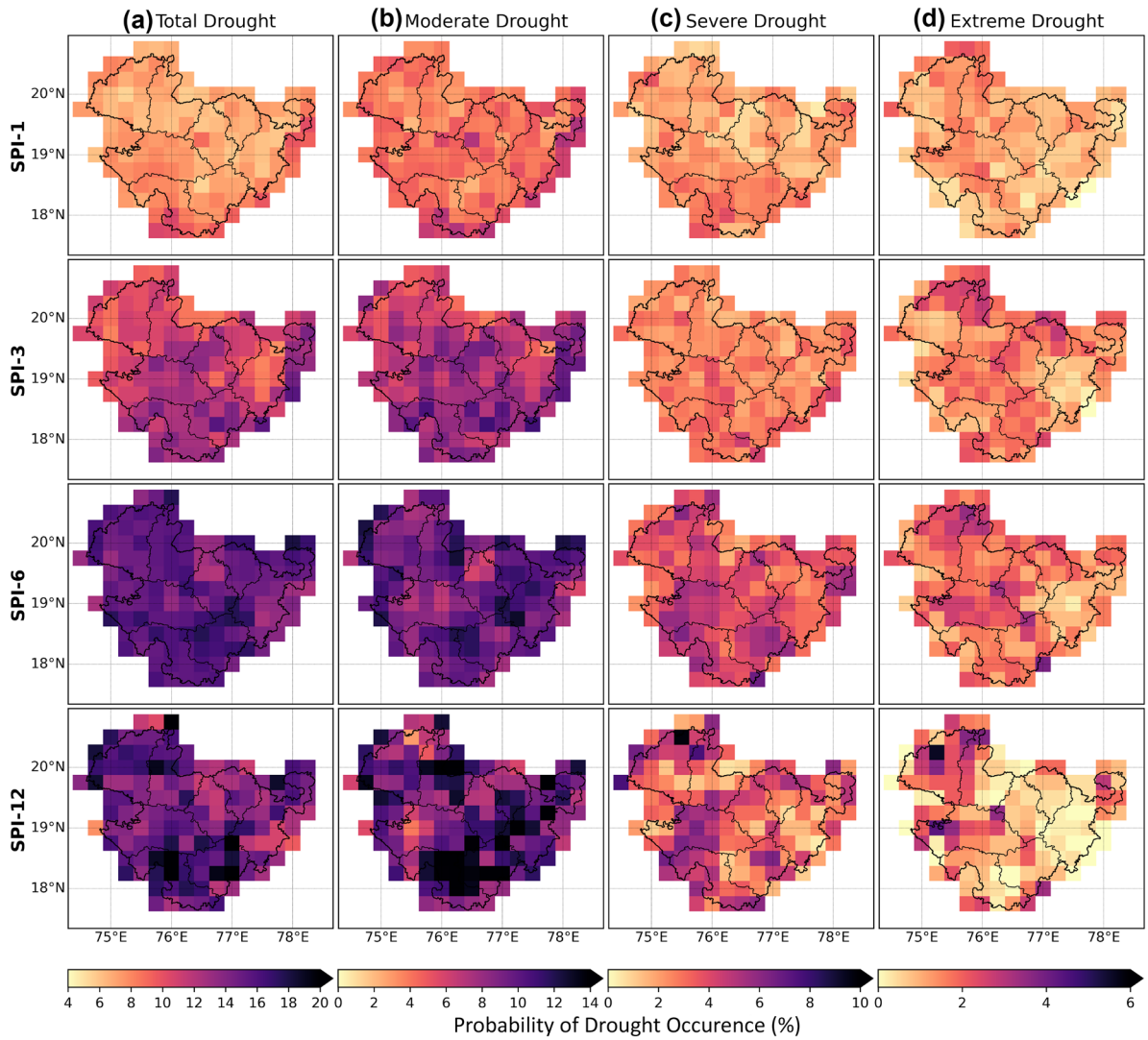



**Fig. 2** Spatial distribution of drought occurrence probability in different moving time frames over the Marathwada Region

high variance/standard deviation of rainfall. It is necessary to understand that SPI conceptually represents the extent of standard deviation ($\sigma$) of a precipitation amount from its mean value ($\mu$), provided the precipitation amounts are normally distributed. The higher values of $\sigma$ lead to failure of SPI to capture drought occurrences at shorter time frames. SPI is highly sensitive to coefficient of variation, which is the ratio of $\sigma$ and $\mu$. This is also the primary cause for the failure of SPI to capture droughts over arid regions, where the coefficient of variation is typically very high. The results are also reflected in the boxplots summarizing the drought occurrence probability in the severity categories and in total over the Marathwada Region at different moving time frames, as shown in Fig. 3. It is obvious that the drought characteristics will vary if the time frame is altered. Nevertheless, the study attempts to justify a better approach based on statistical considerations. The variations and skewness in a time series reduce the power of the statistical tests, especially for trend analysis (Yue et al., 2002). Considering all these aspects, this study recommends that SPI-12 should be preferred for drought monitoring as it provides more consistent information.

The results of the MMK test for trend analysis of SPI series at multiple time frames over the Marathwada Region are presented in Fig. 4. It is worth mentioning that Fig. 4 shows the results only for monsoon months, i.e., June to September. A mix of positive/negative trends of SPI series are found. The trend analysis through the MMK test may get affected by the randomness in the time series, as discussed in previous studies (Swain et al., 2022c; Yue et al., 2002). The power of non-parametric tests is a decreasing function of the variation of the time series. Furthermore, the power of trend tests is dependent on the distribution type, sample size, sample variation, and sample skewness (Yue et al., 2002). The sample distribution is assumed to be inherently normal and therefore, the $Z_{MMK}$ magnitudes of 1.645, 1.96, and 2.576 are taken as the threshold for significance levels of 10%, 5%, and 1%, respectively. In all the time frames (1-, 3-, 6-, and 12-monthly), the sample size is taken the same; however, at a shorter time frame, the sample variations and the sample skewness are higher, thereby reducing the power of the statistical tests. Furthermore, the skewness and fluctuations always affect the normality of the time series. Hence, the results of the MMK test can be considered more reliable at longer time frames.

Given the fact that SPI is the transformed version of rainfall amount, a decreasing trend of SPI represents a decreasing rainfall, and hence, it indicates an increasing or exacerbating drought conditions. More importantly, SPI-1 presents the trend of monthly rainfall for the individual months of the monsoon season, whereas SPI-3, SPI-6, and SPI-12 represent the trend of rainfall accumulations for 3, 6, and 12 months, respectively. From Fig. 4, it is evident that the monthly rainfall (i.e., SPI-1) is

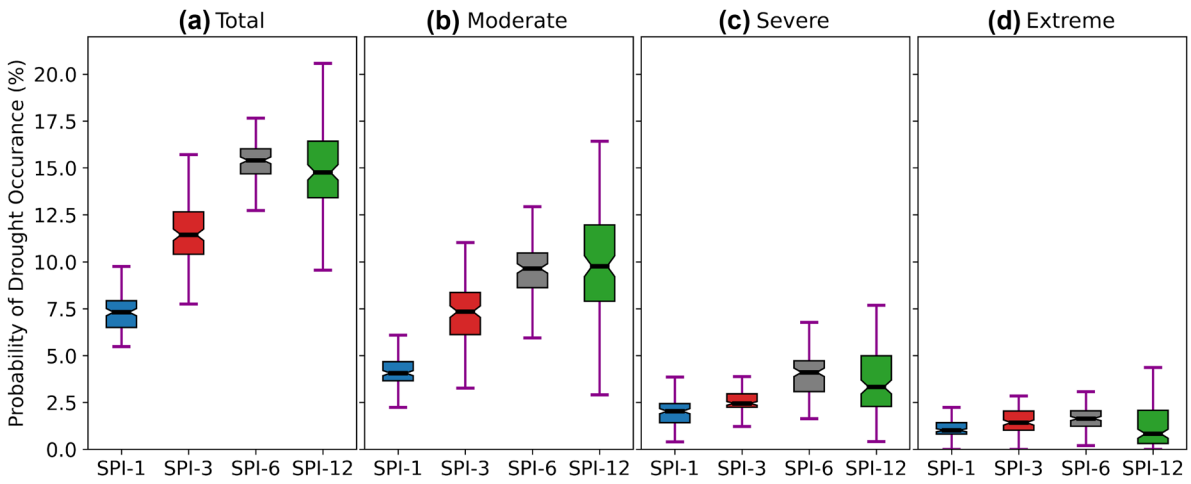



**Fig. 3** Boxplots summarizing the drought occurrence probability in the severity categories and in total over the Marathwada Region at different moving time frames

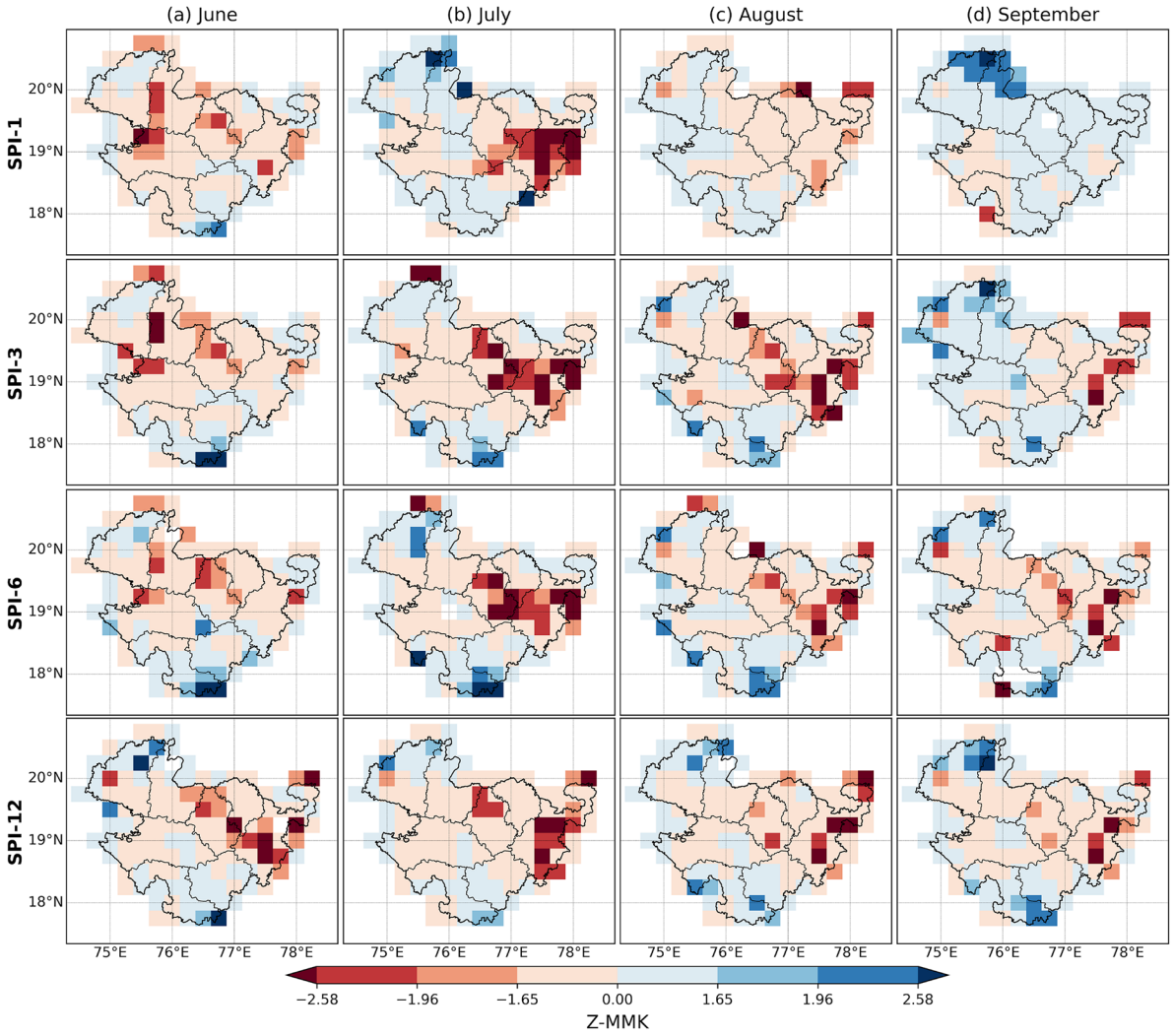



**Fig. 4** Spatial distribution of $Z_{MMK}$ for SPI series in different moving time frames over the Marathwada Region

significantly decreasing in the month of July for some grids in Nanded, Hingoli, and Parbhani districts. Similar results are also noticeable for 3-, 6-, and 12-monthly time frames. More importantly, the difference is the trend statistic (i.e., $Z_{MMK}$) values at 1-, 3-, and 6-monthly time frames depict a significant decrease from June to July. Typically, the effects of a sudden increase/decrease in 1 month reduce with the increase in the time frame. However, the effects of decreasing trend in July for SPI-1 are also reflected in longer time frames, which implies there may be a sudden fall/decrease of rainfall in July. Tacitly, it also indicates the possibility of flash droughts over these regions in July. There is a decreasing trend of SPI or an exacerbating drought severity at all time frames over the Nanded district, in particular. Overall, majority of the grids exhibit non-significant (i.e., $-1.65 < Z_{MMK} < 1.65$) trends. The trend analysis of droughts in historical time series would help better understand the regional drought characteristics.

From the spatial plots of Fig. 4, it may be difficult to discern the exact trend of SPI series over the entire region. This is due to the fact that there are a number of grids over the region, and they may exhibit different results of the MMK test. Therefore, the concentration of grids based on $Z_{MMK}$ for SPI in different moving time frames over the Marathwada Region is summarized through violin plots in Fig. 5. It can be observed

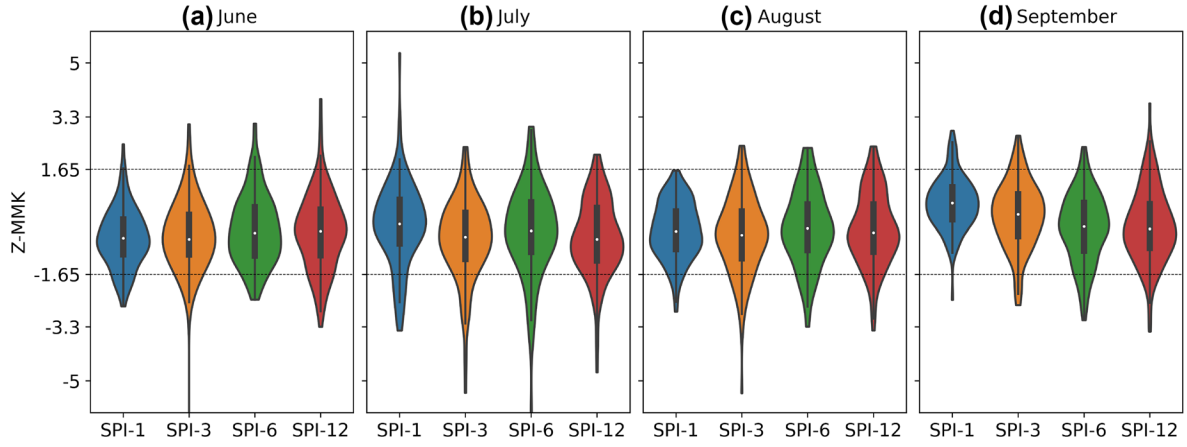



**Fig. 5** Violin plots summarizing the concentration of grids based on $Z_{MMK}$ for SPI in different moving time frames

that while a majority of grids are concentrated in the non-significant ($-1.65 < Z_{MMK} < 1.65$) zone, they are relatively more concentrated towards negative $Z_{MMK}$. For the months of June, July, and August, the concentration/number of grids with a significantly negative trend ($Z_{MMK} < -1.65$) of SPI is higher than that of significantly positive ($Z_{MMK} > 1.65$) ones. Hence, it can be inferred that the Marathwada Region exhibits a relatively increased tendency towards drought occurrences.

To develop a better understanding of meteorological drought trends and variabilities, the magnitudes and trends of meteorological variables, i.e., rainfall, ET, and ESI, are analyzed at seasonal and annual scales. The spatial distribution of mean seasonal/annual rainfall over the Marathwada Region is presented in Fig. 6a. It can be observed that the monsoon season clearly dominates in terms of contribution to annual rainfall, followed by post-monsoon, pre-monsoon, and winter seasons. The mean monsoonal and annual rainfall show remarkable spatial variation and they increase from west to east. The violin plots summarizing the concentration of grids based on mean seasonal/annual rainfall are presented in Fig. 6b. The majority of grids receive a mean monsoonal rainfall in the range of 500–700 mm, whereas the maximum value goes above 1000 mm. Similarly, the maximum value of mean annual rainfall is close to 1200 mm, whereas a majority of the grids receive a mean annual rainfall in the range of 600–800 mm. Hence, the monsoon season contributes about 85% of the annual rainfall over the study area.

The results of trend analysis of rainfall at seasonal and annual scales over the Marathwada Region are presented in Fig. 7. The spatial distribution of $Z_{MMK}$ for seasonal and annual rainfall is presented in Fig. 7a, whereas Fig. 7b presents the concentration of grids based on $Z_{MMK}$ for seasonal/annual rainfall. The trend of rainfall is mostly increasing for pre-monsoon season with a significantly positive trend over some grids, whereas all the grids showed a non-significant trend during post-monsoon season. The results of the winter season seem opposite to that of the pre-monsoon season; i.e., the trend is mostly decreasing, and significantly decreasing over some grids. However, these seasons cumulatively contribute about 15% of annual rainfall, and hence, the trend of rainfall in the monsoon season governs the trend of annual rainfall. A mix of positive and negative trends of rainfall is observed for the monsoon season, and most of them were non-significant. For annual rainfall, the results are similar; however, the concentration of grids is relatively higher towards negative $Z_{MMK}$. Moreover, the number/concentration of grids possessing significantly decreasing trends is slightly higher than the ones with significantly increasing trends.

The spatial distribution of RC obtained from the SS test for seasonal and annual rainfall is presented in Fig. 7c, whereas Fig. 7d presents the concentration of grids based on RC for seasonal/annual rainfall. The results are consistent with that of the MMK test. Interestingly, the RC values are very high for non-monsoon seasons. RC represents the magnitude of

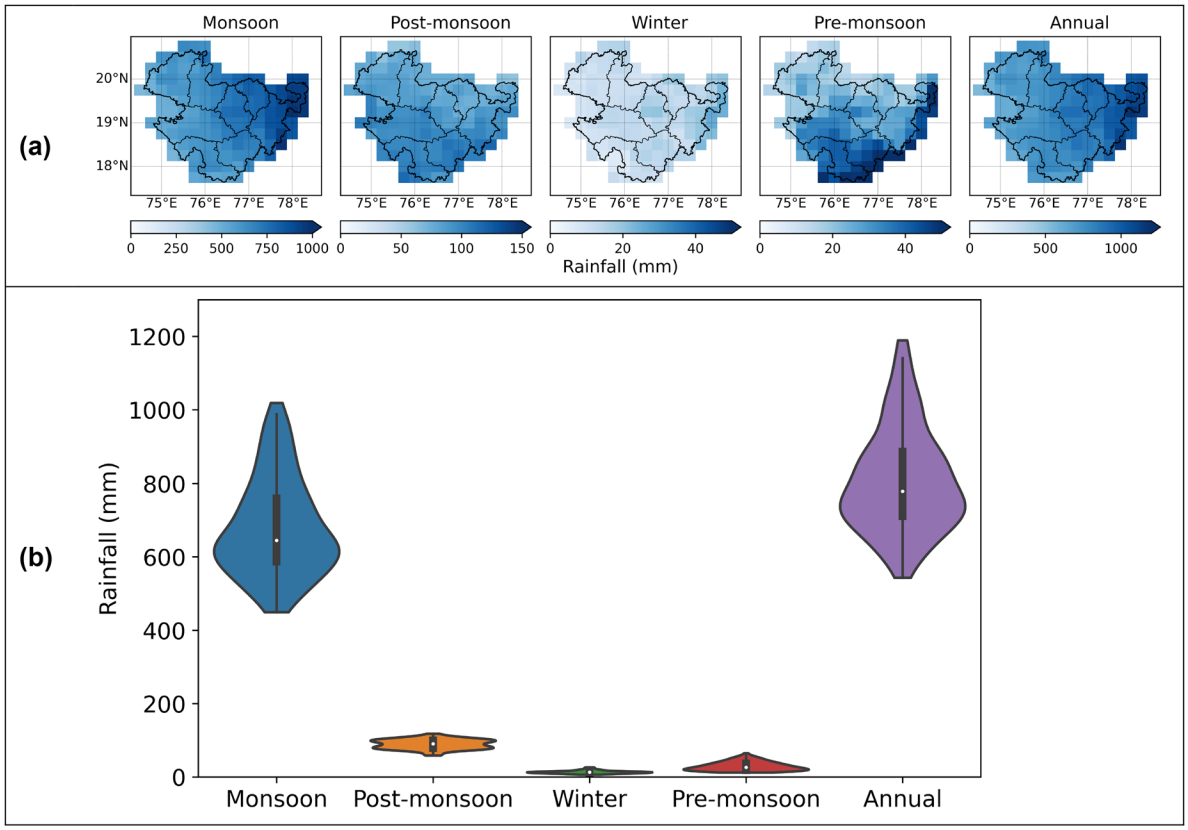



**Fig. 6** **a** Spatial distribution of mean seasonal/annual rainfall over the Marathwada Region. **b** Violin plots summarizing the concentration of grids based on mean seasonal/annual rainfall

changes with respect to the long-term mean value. As the mean rainfall in these seasons is scanty, a slight change appears to be larger with respect to the mean value. Furthermore, RC is obtained from Sen's slope, which is the median of slopes for all pairs of ordered (ordinal) time points. Therefore, if there are a lot of zero values in data points, the median slope turns out to be zero. This is the reason the months or seasons receiving scanty rainfall exhibit zero value of Sen's slope and RC. The higher concentration of grids towards negative RC for monsoonal and annual rainfall is clearly observable from Fig. 7d.

The spatial distribution of mean seasonal/annual ET over the Marathwada Region is presented in Fig. 8a, whereas the violin plots summarizing the concentration of grids based on mean seasonal/annual ET are presented in Fig. 8b. Similar to rainfall, the monsoon season clearly dominates in terms of contribution to the annual ET, followed by post-monsoon, winter, and pre-monsoon seasons. However, in contrast to rainfall, the spatial variation in the annual ET is not very high. The majority of the grids receive a mean monsoonal ET in the range of 250–280 mm, whereas the maximum value goes close to 300 mm. Similarly, the maximum value of the mean annual ET exceeds 650 mm; however, a majority of the grids receive a mean annual ET in the range of 530–570 mm. Hence, the monsoon season contributes about 50% of the annual ET over the Marathwada Region. It should be noted that ET is a continuous process, and its magnitude depends primarily on water available in the land (in the form of soil moisture) and water bodies. In the context of Marathwada, monsoon is the wettest season, which subsequently gets drier towards post-monsoon and winter and reaches the driest condition in pre-monsoon season. Therefore, ET is maximum in monsoon season and minimum in pre-monsoon or summer season.

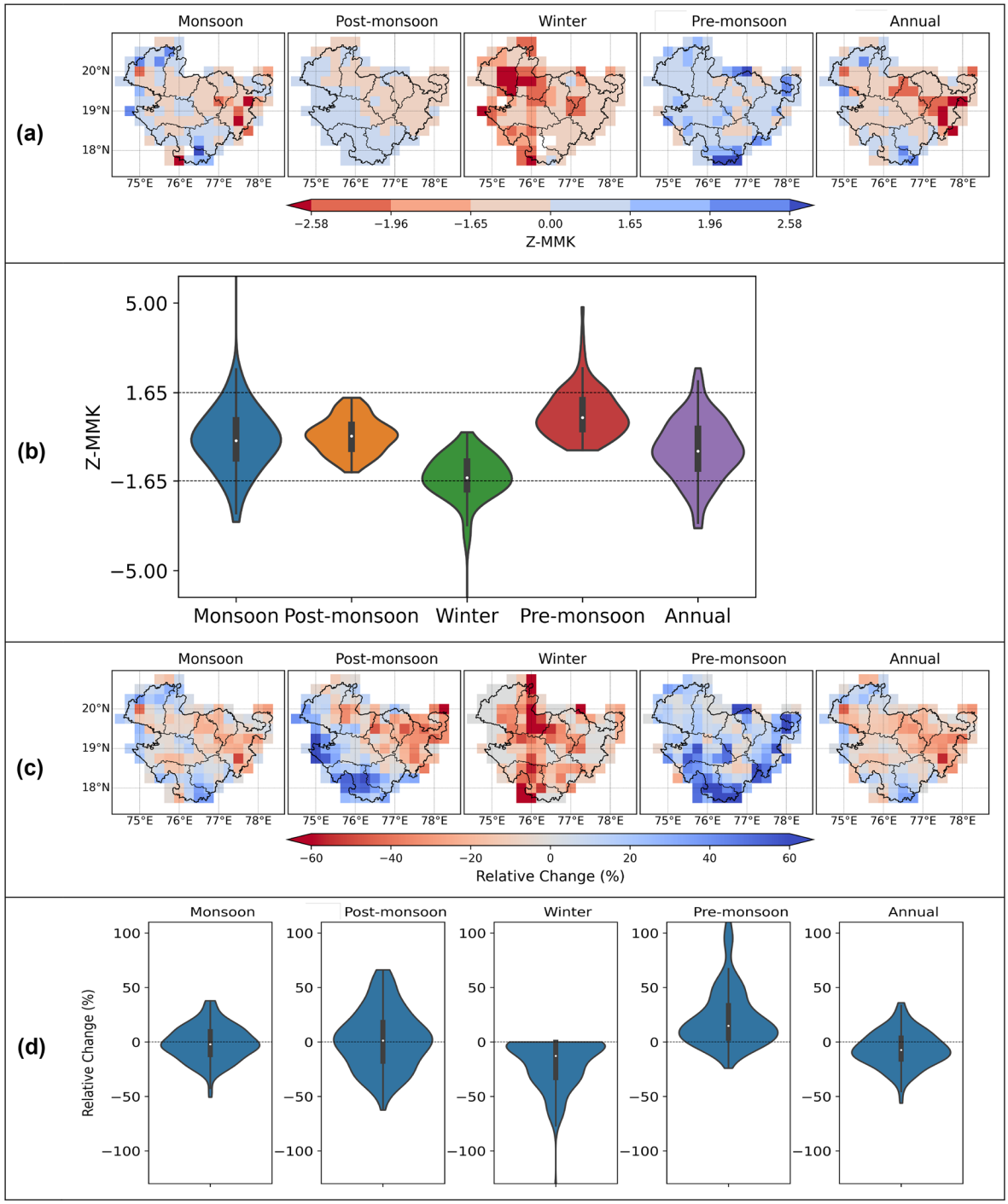



**Fig. 7** **a** Spatial distribution of $Z_{MMK}$ for seasonal/annual rainfall. **b** Concentration of grids based on $Z_{MMK}$ for seasonal/annual rainfall. **c** Spatial distribution of RC for seasonal/annual rainfall. **d** Concentration of grids based on RC for seasonal/annual rainfall

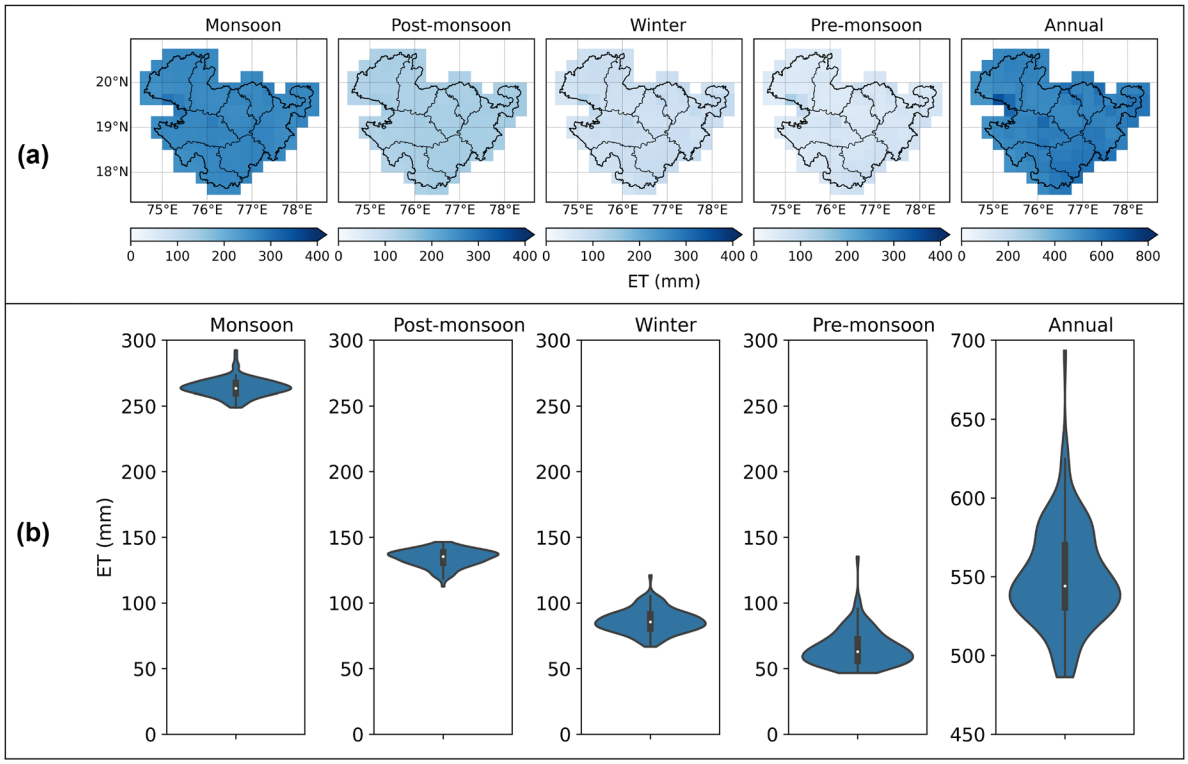



**Fig. 8** **a** Spatial distribution of mean seasonal/annual ET over the Marathwada Region. **b** Violin plots summarizing the concentration of grids based on mean seasonal/annual ET

The results of trend analysis of ET at seasonal and annual scales over the Marathwada Region are presented in Fig. 9. The spatial distribution of $Z_{MMK}$ for seasonal and annual ET is presented in Fig. 9a, whereas Fig. 9b presents the concentration of grids based on $Z_{MMK}$ for seasonal/annual ET. The trend of ET is mostly non-significant (significantly positive for a few grids) for monsoon season. On the other hand, almost all the grids showed a positive trend of ET in post-monsoon, winter, and pre-monsoon seasons. More importantly, a majority of them are significant. It can be observed from Fig. 9b that all the grids showed significantly increasing trends of ET during the pre-monsoon season. The annual ET is also significantly increasing over almost the entire study area.

The spatial distribution of RC obtained from the SS test for seasonal and annual ET is presented in Fig. 9c, whereas Fig. 9d presents the concentration of grids based on RC for seasonal/annual ET. The results are consistent with that of the MMK test. Most of the grids show an increase of 20% with respect to the mean annual ET. The increasing ET can be attributed to the increasing temperature. Climate change and global warming leading to a remarkable increase in temperature would result in higher evapotranspiration. This is a matter of concern for the agriculture-dominated Marathwada Region, as it challenges fulfilling the crop water requirements. Besides, the increase in irrigation might have contributed to the increase in ET during the non-monsoon seasons.

The spatial distribution of mean seasonal/annual ESI over the Marathwada Region is presented in Fig. 10a, whereas the violin plots summarizing the concentration of grids based on mean seasonal/annual ESI are presented in Fig. 10b. Unlike rainfall and ET, the ESI during monsoon and post-monsoon seasons are quantitatively similar, whereas it is lowest in the pre-monsoon season. However, like ET, the spatial variation in the annual ESI is not very high. The majority of the grids possess a mean ESI above 0.8 during monsoon and post-monsoon

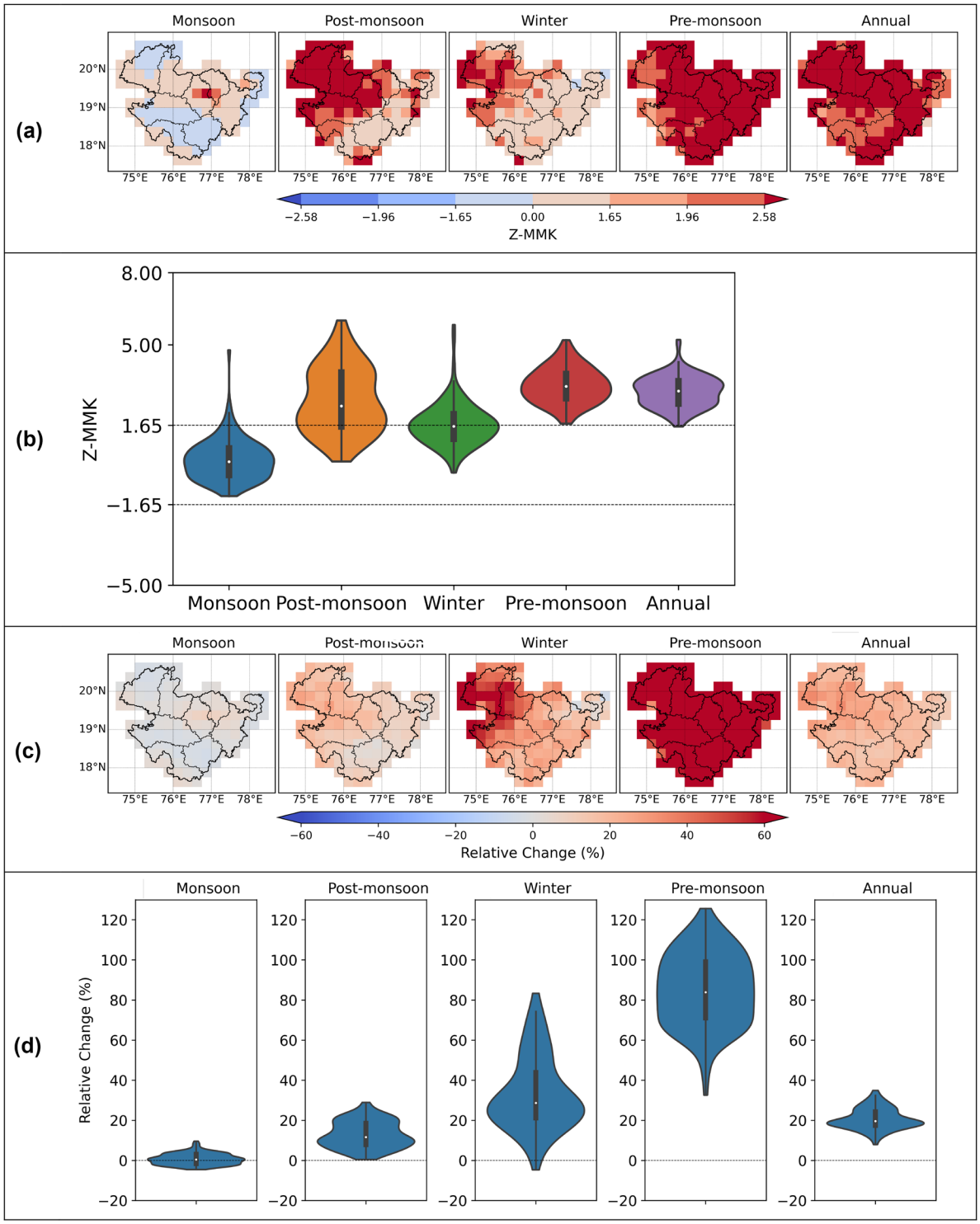



**Fig. 9** **a** Spatial distribution of $Z_{MMK}$ for seasonal/annual ET. **b** Concentration of grids based on $Z_{MMK}$ for seasonal/annual ET. **c** Spatial distribution of RC for seasonal/annual ET. **d** Concentration of grids based on RC for seasonal/annual ET

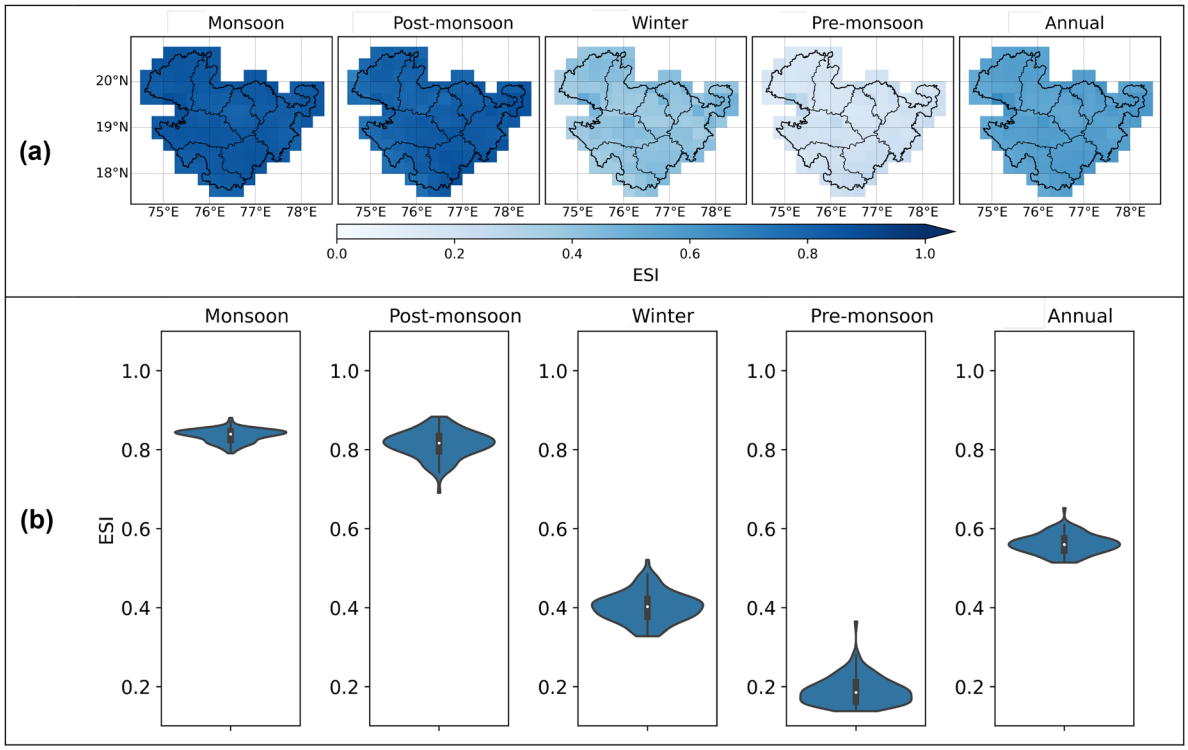



**Fig. 10** **a** Spatial distribution of mean seasonal/annual ESI over the Marathwada Region. **b** Violin plots summarizing the concentration of grids based on mean seasonal/annual ESI

seasons. On the other hand, the mean ESI over winter and pre-monsoon seasons are concentrated around 0.4 and 0.2, respectively. As ESI is computed as ET/PET, the seasonal differences are explicable in terms of ET and PET. Though ET in monsoon is higher compared to that in post-monsoon (Fig. 8b), the temperature in the latter is also relatively lower, which leads to a lower mean PET. This leads to closer values of the mean ESI in both seasons. Similarly, the pre-monsoon season witnesses the lowest ET and the highest PET among all the seasons, leading to the lowest mean value of ESI. For winter season, the temperature and, consequently, the PET is the lowest. This along with a relatively higher ET (compared to pre-monsoon season), leads to a higher mean ESI. Moreover, as the study area is majorly dominated by agricultural lands, the irrigation during the winter season to meet the water requirements of Rabi crops may contribute to a higher ET. The annual mean ESI is mostly concentrated between 0.5 and 0.6.

The results of trend analysis of ESI at seasonal and annual scales over the Marathwada Region are presented in Fig. 11. The spatial distribution of $Z_{MMK}$ for mean seasonal and annual ESI is presented in Fig. 11a, whereas Fig. 11b presents the concentration of grids based on $Z_{MMK}$ for mean seasonal/annual ESI. It is evident that the trend of ESI is significantly positive for most of the grids during monsoon, post-monsoon, and pre-monsoon seasons. For winter season, almost all the grids also showed positive values of $Z_{MMK}$, and the concentration is higher at non-significant trends ($Z_{MMK} <$ 1.65). The annual mean ESI is also significantly increasing over almost the entire study area. The results can be attributed to the increasing trends of ET. The increase in evaporative stress should be considered seriously from the point of view of agricultural water management. The increase in ET and ESI should be considered an alarm, as these would further exacerbate the drought severities, particularly by drastically reducing the available soil moisture for crops. The results also necessitate the development of proper action plans to combat

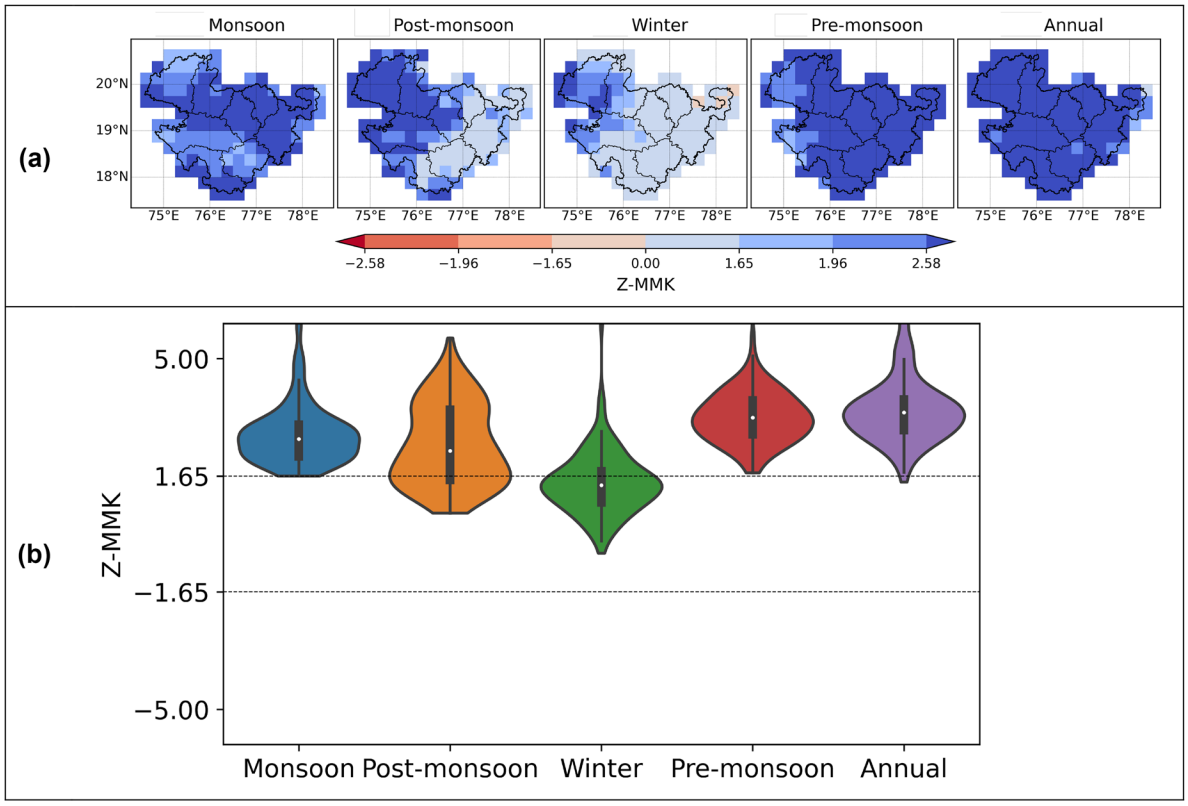



**Fig. 11** **a** Spatial distribution of $Z_{MMK}$ for seasonal/annual ESI. **b** Concentration of grids based on $Z_{MMK}$ for seasonal/annual ESI

the impacts of severe droughts. Furthermore, considering the trends of droughts and meteorological variables, this study recommends the cultivation of crops with low water requirements.

## Conclusions

A detailed spatiotemporal assessment of drought trends and variabilities during 1980–2020 over the Marathwada Region, Maharashtra, India, is carried out using statistical approaches. The region is found to be prone to droughts. SPI at longer time frame (i.e., 12-monthly moving frame) is found to capture drought occurrences better than the shorter time frames, which can be attributed to the lesser randomness in the time series at a longer frame. The trend analysis of SPI series for the monsoonal months revealed a mix of positive and negative trends over different grids; however, they are relatively more concentrated towards negative $Z_{MMK}$, which indicate a relatively increased tendency towards drought occurrences or increased drought severities during the studied duration. The seasonal differences in mean values and trends of rainfall, ET, and ESI are discussed in detail. The information reported in this study will help in devising water management strategies to minimize the repercussions of droughts over the agriculture-dominated Marathwada Region. Besides, the methodology presented will encourage its replication over different regions of the world that have been affected by climate change and its ramifications, mostly in the form of frequent droughts or lower crop yields.

**Author contribution** Sabyasachi Swain: conceptualization, data curation, formal analysis, visualization, writing—original draft. Surendra Kumar Mishra: supervision, review, and editing. Ashish Pandey: supervision, review, and editing. Deen Dayal: data curation, visualization.

**Availability of data and material** The gridded rainfall data from IMD is freely available all over India. The ET and PET data are extracted from GLEAMS, which is an open-access global dataset. The detailed information on their sources is provided inside the manuscript.

**Code availability** The Python codes are available from the corresponding author upon reasonable request. The Python library 'IMDLIB' (https://doi.org/10.5281/zenodo.4405233) can be used to read and download the IMD gridded data. More details on IMDLIB can be found here: https://imdlib.readthedocs.io/en/latest/

## Declarations

**Conflict of interest** The authors declare no competing interests.

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